

Title : **Natural Language Explainable CKD Prediction via Fine-Tuned LLMs**

1. Introduction :

Chronic Kidney Disease (CKD) is a progressive decline in kidney function that often goes undetected until its later stages, reducing filtration rate and ultimately leading to the need for dialysis or transplantation [1,2]. It is a major global health burden, affecting roughly 10% of adults worldwide and more than 800 million people overall [2], putting a heavy burden on healthcare systems due to its high prevalence and costly treatments. Early detection and identification of this disease are crucial for enhancing patient outcomes [3] and alleviating the burden on healthcare systems. Traditional diagnostic approaches, primarily the glomerular filtration rate (GFR) and urine tests for albumin, are useful for diagnosing and monitoring Chronic Kidney Disease (CKD) but remain ineffective in accurately predicting its progression or identifying the most significant factors contributing to its development [3].

Recent studies demonstrate that supervised Machine Learning (ML) and Deep Learning (DL) models trained on structured clinical data can predict CKD presence, stage, and progression with remarkable accuracy, often reaching the 99% mark [4,5,6]. Tree-based machine learning methods, including random forest, XGBoost, and gradient boosting, as well as neural network models, consistently demonstrate strong predictive performance across studies and emerge as top performers [1].

Despite these advances, most of these models still behave as ‘black boxes’, as they are hard to understand and provide almost no information on why a prediction is made [2]. Clinicians need not only accurate risk scores but also transparent, patient-specific explanations that facilitate decision-making and help identify potential model errors or biases [7,8]. In response, recent studies have increasingly integrated predictive modeling with explainable artificial intelligence (XAI) techniques such as Shapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and fuzzy rule-based systems to highlight the contribution of individual features behind each prediction, thereby providing transparent outputs that can support clinical decision-making [2,7,8].

However, still several challenges still remain. According to this study [9], XAI methods provide valuable insights into feature influence; primarily offering technical attributions and plots that require specialized expertise to interpret and do not directly deliver intuitive, human-centered rationales for individual predictions. In contrast, rationalization aims to justify model outputs through natural-language explanations that explicitly articulate the reasoning behind a prediction. Such rationales are inherently human-comprehensible and accessible to non-technical users, making them particularly valuable in clinical settings where transparency and trust are essential.

In parallel, Large Language Models (LLMs) have emerged as powerful tools capable of generating human-like, natural language text responses (rationales) to diverse user queries [10], and application of these models is currently being explored on tabular data as well [11,12,13]. To directly address the need for user-friendly and justifiable explanations mentioned in these studies

[1,9], we propose a novel methodology that leverages the Natural Language Generation (NLG) capabilities of Large Language Models (LLMs) to create intuitive, easy-to-understand rationales for CKD prediction while supporting the need for AI integration in kidney disease care.

Specifically, the objectives of this work are:

1. To propose a novel approach that generates case-specific natural language rationales explaining each CKD prediction;
2. To benchmark large language models (LLMs) against traditional machine learning models for CKD prediction on clinical data.

Organization of paper to be included.

2. PREVIOUS WORK:

Large Language Models (LLMs) have recently gained prominence in healthcare due to their ability to process complex, context-rich information and support disease prediction while offering evidence-based explanations [11], [17]. Major technology leaders have introduced medical-oriented LLMs—such as Med-PaLM 2 and BioGPT—recognized for their strong summarization abilities and their capacity to generate clear and interpretable reasoning alongside diagnostic predictions [6], [12], [8]. Recent studies further demonstrate that general-purpose LLMs can achieve competitive performance relative to traditional machine learning and deep learning models on tabular classification tasks, including those involving healthcare data [7]. This research aims to benchmark the performance of fine-tuned LLMs against traditional ML models for chronic kidney disease (CKD) prediction, with an emphasis on evaluating interpretability. Addressing concerns highlighted regarding the lack of transparency in “black-box” models [10], this study adopts a self-explanation prompting strategy, enabling the LLM to generate natural-language reasoning for each prediction. This approach enhances transparency and trust without the need for post-hoc interpretability techniques such as SHAP or LIME.

3. Methodology :

The section describes the proposed work. It begins by explaining how the data was collected and transformed. Next, it presents benchmarking results comparing the Large Language Model with traditional Machine Learning models. Finally, it outlines the approach used to generate natural language responses for each prediction.

3.1. Data Collection

Three CKD-related datasets were employed in this research , comprising of individuals with and without chronic kidney disease. These datasets were sourced from clinical centers and opensource dataset repositories. By utilizing multiple datasets , we aimed to test whether the approach generalizes well on diverse datasets rather than reporting results from only one dataset. Collectively these datasets encompasses :

- (i) demographic variables such as age, gender, and ethnicity

- (ii) laboratory measurements including serum creatinine and estimated glomerular filtration rate (eGFR)
- (iii) clinical observations such as blood pressure, diabetes status, and reported symptoms.

3.1.1. Data Sources

- Details about the datasets is provided below :
 - **Pakistan Kidney Center (PKC) Dataset (pkc):** This is a region specific dataset collected form Pakistan Kidney Center , Abbotabad. It is not publicly available.
 - **UCI Chronic Kidney Disease Dataset (uci):** This dataset was downloaded from University of California Irvine Machine Learning Repository and is publicly available.
 - **Spanish CKD Dataset (spn):** This dataset was downloaded form Kaggle . It is named as Spanish because it was in Spanish Language and was later translated to English. It is also publicly available under the title “**Dataset_insuficiencia_renal_crnica**” on Kaggle.

Further detail about the datasets is provided in provided in Table 1 below.

Dataset Name	Total Features	Categorical	Numerical	Shape (rows,cols)
PKC CKD Dataset	22	2	20	398 , 22
UCI CKD Dataset	25	11	14	400 , 25
Spanish CKD Dataset	20	6	14	15050 ,20

These datasets contained variety of features but there were some features that were common in all three datasets which are shown in Table ii.

Common Feature	Column Names Used in Datasets	Data Type
Age	(Age, age)	Numerical
Blood Urea	(Blood Urea, bu, Urea)	Numerical
Serum Creatinine	(Serum Creatinine, sc, Creatinine)	Numerical
Hemoglobin	(Hemoglobin, hemo, Haemoglobin)	Numerical
PCV / Hematocrit	(Packed Cell Volume, pcv, Hematocrit)	Numerical
WBC Count	(White Blood Cell Count, wc, WBC Count)	Numerical
RBC Count	(Red Blood Cell Count, rc, RBC Count)	Numerical

Diagnosis/Target	(Diagnosis, classification, Target)	Categorical
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3.1.2. Data Preprocessing

- **Data Cleaning** : UCI and Spanish Dataset both contained null values which can be seen in the the figure below while PKC Dataset was clean with no null values. So dropping and imputation techniques were used to treat nullity in columns.
- **Outlier Removal** : Spanish CKD Dataset contained lots of outliers, so Interquartile Range (IQR) method was used to identify and remove those values. For each numerical column x , the first and third (Q_1 , Q_3) quartiles were computed and IQR range was defined as :

$$IQR = Q_3 - Q_1$$

Lower and upper bounds were then set to $L = Q_1 - 1.5 \cdot IQR$ and $U = Q_3 + 1.5 \cdot IQR$;

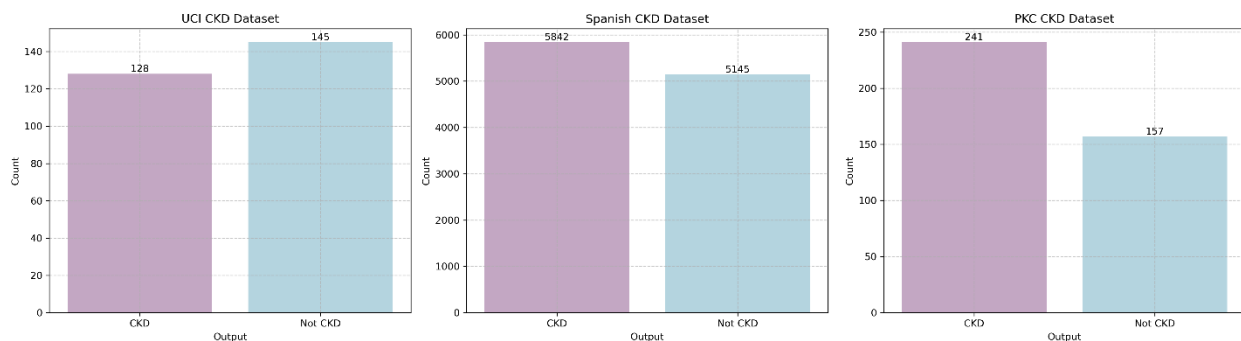
any observation with $x < L$ or $x > U$ was treated as an outlier and removed from the dataset prior to model training.

- **Data Partitioning** : Each dataset was splitted into 80 / 20 ratio for model development and testing.

So after implementing data preprocessing on datasets , Table iii shows how they transformed from Table i.

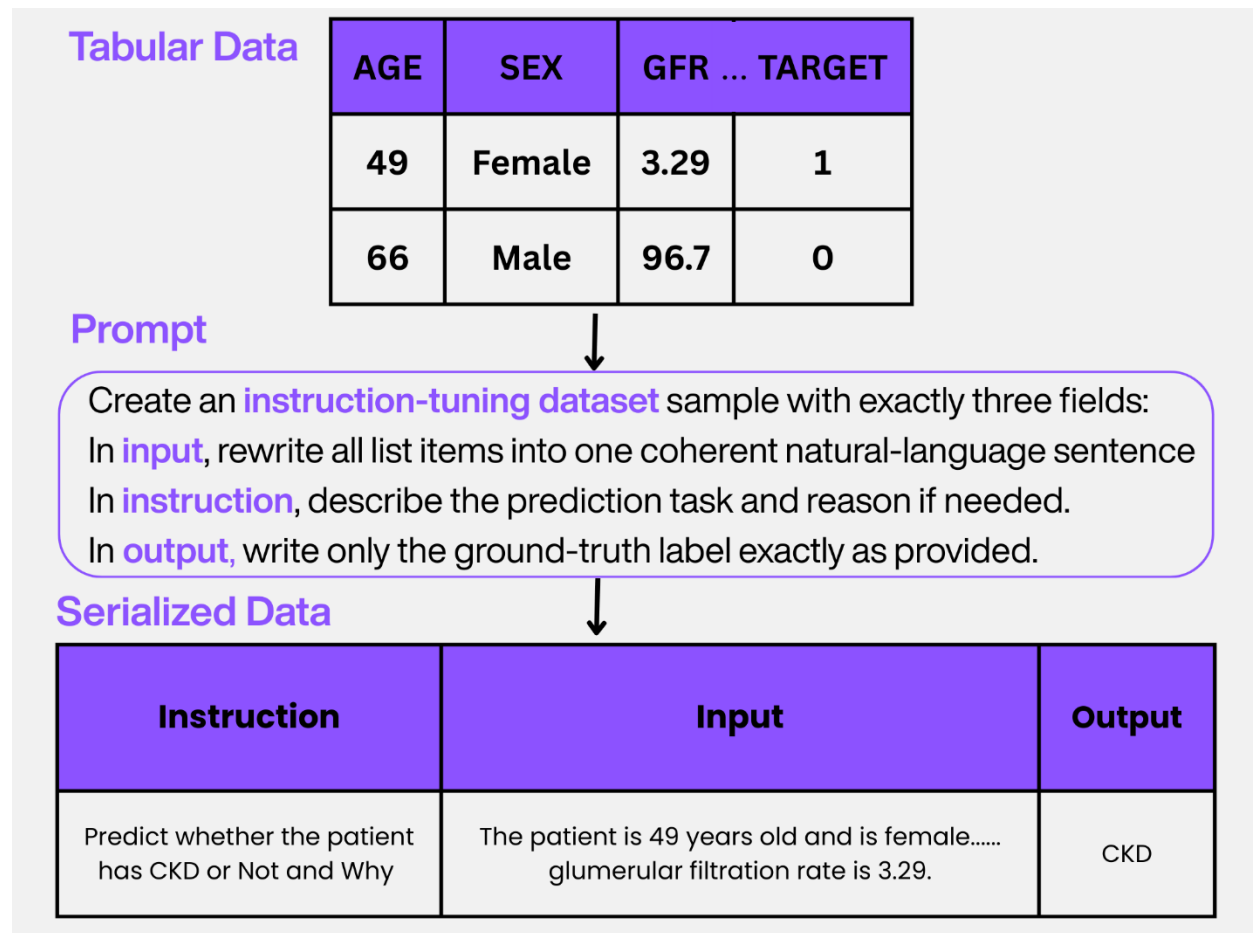
Dataset Name	Total Features	Categorical	Numerical	Shape (rows,cols)
PKC CKD Dataset	22	2	20	398 , 22
UCI CKD Dataset	25	11	14	273 , 25
Spanish CKD Dataset	20	6	14	10987 , 20

Figure i illustrates the distribution of target classes following data preprocessing.



C. Data Serialization for LLMs:

We are using LLM to perform predictions on clinical data, but it cannot process tabular data, just like traditional Machine Learning Models. To make LLM process tabular data, the table must be transformed into natural language text. We used **Text GPT3** method to serialize the tabular data into natural language text as mentioned in [11]. Figure ii illustrates the transformation of each tabular patient record into an LLM-readable textual sequence via **serialization**, following an **instruction-tuning** format (instruction, input, output) used for model training/inference.



D. Model Training and Evaluation :

This section describes the predictive modeling strategies along with explainability of prediction employed in this study. We used llm finetuning and traditional ml model training approach to compare the results. Both models were trained and evaluated on identical datasets to ensure a fair and consistent benchmarking protocol.

Large Language Model Fine-Tuning on Serialized Data:

A pre-trained Large Language Model (LLM) was employed to be finetuned for CKD prediction and explanation. Specifically, the **Gemma-3N-E4B-it** model which is a **1b** parameter variant of

Gemma-3N-E4B was selected due to its small size and efficient performance as we had limited compute to conduct this experiment.

3.1.1 Parameter-Efficient Fine-Tuning (PEFT)

To reduce computational cost and mitigate overfitting, **Parameter-Efficient Fine-Tuning (PEFT)** was employed using **Low-Rank Adaptation (LoRA)**. LoRA works by keeping base model parameters frozen, and introducing only a small set of trainable low-rank adapter matrices into the attention, and feed-forward layers of the model. Additionally, 4-bit quantization was applied to the base model weights (QLoRA), significantly reducing memory and computational requirements with negligible impact on model performance. The model was fine-tuned using supervised instruction tuning with gradient accumulation to enable a larger batch size.

During supervised fine-tuning, we configured the training loop with a **maximum of 60 steps** for the PKC and UCI tasks, and **100 steps for the SPN task** (max_steps). We used a **learning rate of 2×10^{-4}** , which strikes a balance between stable convergence and effective adaptation. To support larger effective batch sizes without exceeding memory limits, we applied **gradient accumulation with 4 steps** (gradient_accumulation_steps = 4).

For optimization, we selected **8-bit AdamW** (optim = "adamw_8bit") with a **weight decay of 0.01** to regularize the model and help mitigate overfitting. Within LoRA, we set the **rank (r) to 8** and **LoRA alpha to 8**, which controls the expressiveness and scaling of the low-rank adapter matrices inserted into the attention and feed-forward layers.

These hyperparameters, in conjunction with QLoRA quantization and frozen base weights, enabled efficient fine-tuning that preserved performance while greatly reducing compute and memory demands.

Machine Learning Model Training :

Evaluation Metrics :

The predictive performance is quantified using the following mathematical formulations, where TP , TN , FP , and FN represent True Positives, True Negatives, False Positives, and False Negatives, respectively:

Metric	Formula	Description
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	The overall proportion of correct CKD predictions
Precision	$\frac{TP}{TP + FP}$	The ability of the model to avoid false positives in diagnosis.

Recall	$\frac{TP}{TP + FN}$	The sensitivity of the model in identifying all actual CKD cases
F1-Score	$2 \times \frac{Precision \times Recall}{Precision + Recall}$	The harmonic mean of precision and recall for balanced evaluation
ROC-AUC	$\int_0^1 TPR(FPR^{-1}(u))du$	Measures the model's ability to distinguish between CKD and non-CKD classes across all thresholds

Algorithm: Fine-Tuning Medical LLMs for CKD Prediction

Input:

- Heterogeneous Datasets $\mathcal{D} = \{D_{PKC}, D_{UCI}, D_{Ext}\}$
- Pre-trained Medical LLM $\mathcal{M}$ with initial weights W_0
- Training Hyperparameters: Rank r , scaling α , learning rate η

Output:

- Fine-tuned Model $\mathcal{M}_{FT}$
- Predictive Rationale $\mathcal{R}$
- Evaluation Vector $\mathcal{V} = [Acc, Prec, Rec, F1, AUC]$

Step 1: Data Synthesis and Preprocessing

1. Iterate through each $D_i \in \mathcal{D}$:
 - Apply cleaning function $f_{clean}(x)$ to rectify null values and outliers.
 - Perform feature selection via correlation analysis: $X' = \{x \in X \mid \rho(x, y) > \tau\}$.
2. Partition the unified dataset $\mathcal{D}_{final}$ into subsets $S_{train}, S_{val}, S_{test}$ using an 80/20 split.

Step 2: Natural Language Serialization

1. Define template $\mathcal{T}$ for tabular-to-text conversion.
2. For each clinical record $r_j \in \mathcal{D}_{final}$:
 - Generate narrative $N_j = \mathcal{T}(r_{demographics}, r_{labs}, r_{clinical})$.
 - Result: $S_{serialized} = \{(N_j, y_j)\}_{j=1}^n$.

Step 3: PEFT via QLoRA (Unsloth Implementation)

1. Quantize weights W_0 using 4-bit NormalFloat (NF4): $W_{NF4} = Q(W_0)$.

2. Initialize low-rank adapter matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$.
3. Execute forward pass during fine-tuning:

$$h = W_{NF4}x + (BA)x$$
4. Update only parameters in $\Theta = \{A, B\}$ using optimized gradient descent via Unsloth kernels.

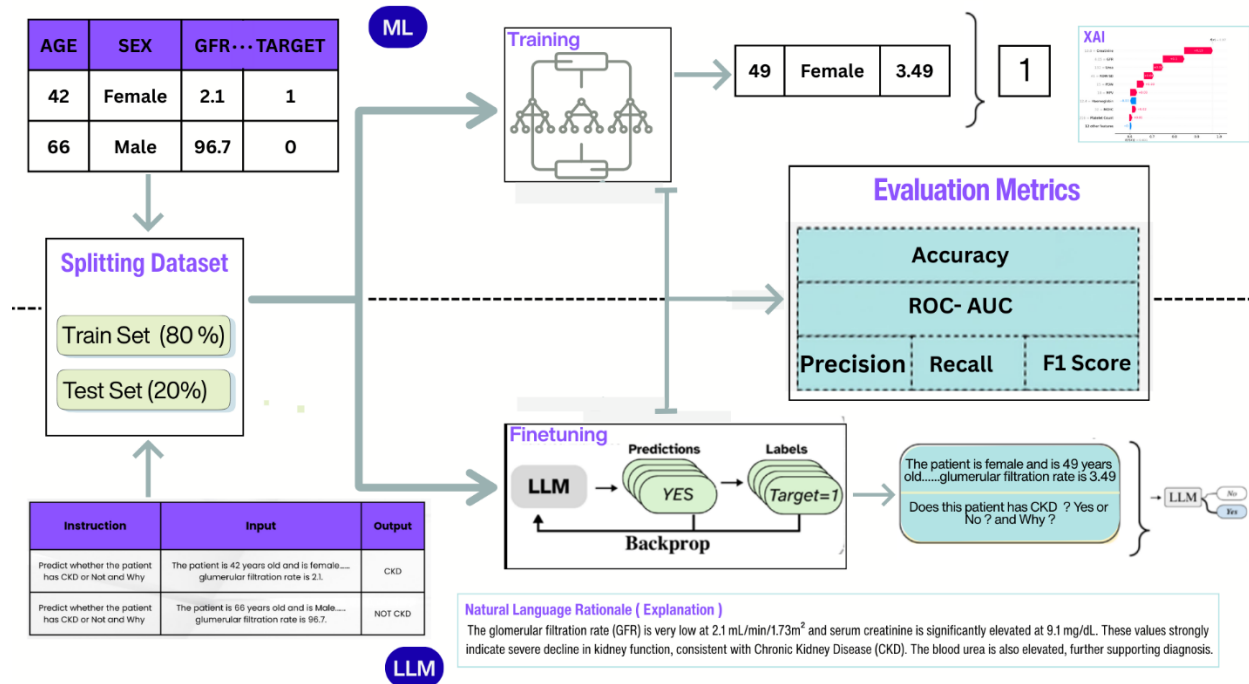
Step 4: Benchmarking and Evaluation

1. Compute the confusion matrix components: TP, TN, FP, FN .
2. Calculate evaluation vector $\mathcal{V}$ using:
 - $Acc = \frac{TP+TN}{Total}$.
 - $F1 = \frac{2 \cdot Prec \cdot Rec}{Prec+Rec}$. etc.
3. Generate Rationale $\mathcal{R}$ via self-explanation prompting: $\mathcal{R} = \mathcal{M}_{FT}(N_{test}, \text{"explain your reasoning"})$.

Step 5: Clinical Deployment

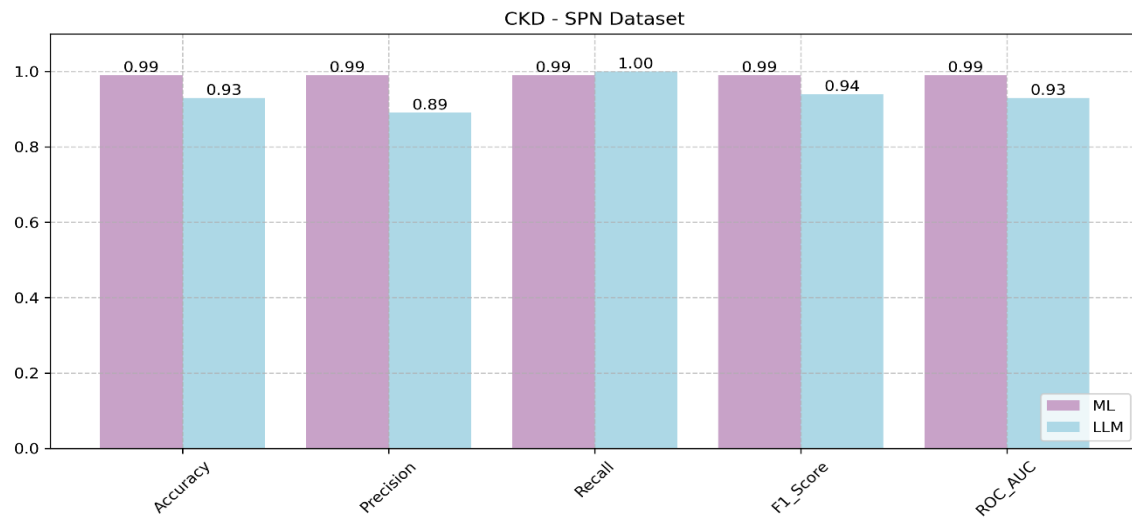
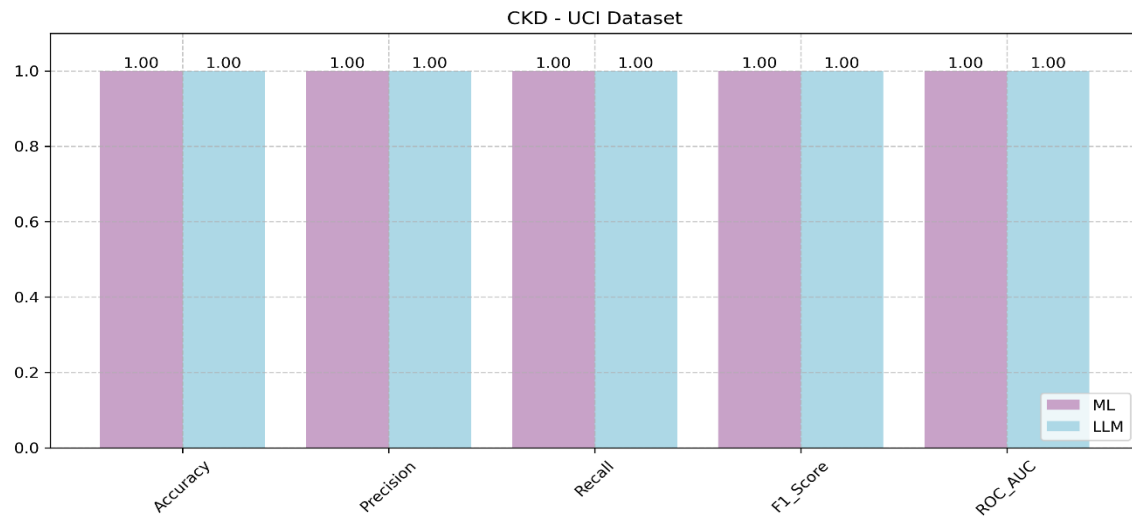
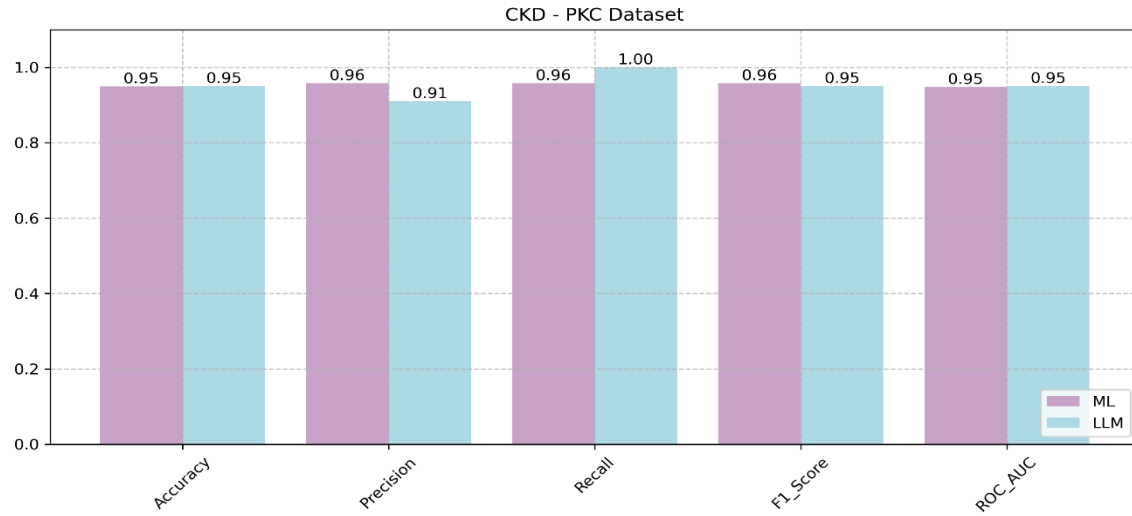
1. Integrate $\mathcal{M}_{FT}$ into the Deployment Layer.
2. Output interpretable risk assessments for clinicians at Pakistan Kidney Center.

E. Deployment and Application Layer : The resulting CKD-specific LLM will be deployed at the Pakistan Kidney Center to support clinicians in early risk assessment and decision-making. Additionally, a user-friendly mobile and web application is being developed to allow individuals to enter personal health parameters and receive instant CKD risk assessments with transparent explanations. This dual deployment approach ensures both clinical impact and patient-centered accessibility.



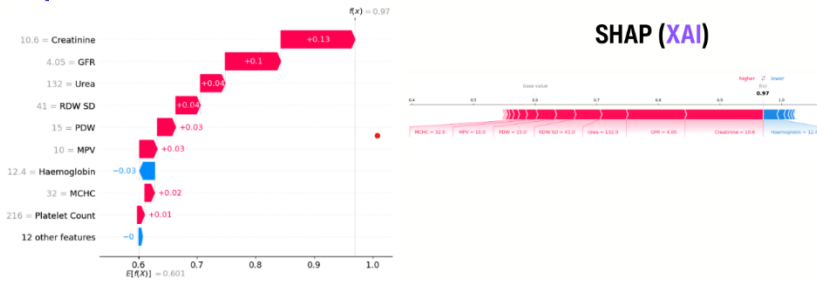
Figures 1 and 2 illustrate the data serialization pipeline and the end-to-end fine-tuning and inference workflow, respectively. The final fine-tuned model generates interpretable CKD risk predictions by extracting key reasoning patterns from serialized patient descriptions.

Results :



Input : {'Sex': 0, 'Age': 39.0, ..., 'Creatinine': 10.6, 'Urea': 132.0, 'GFR': 4.05} (CKD Case)

LLM Rationale

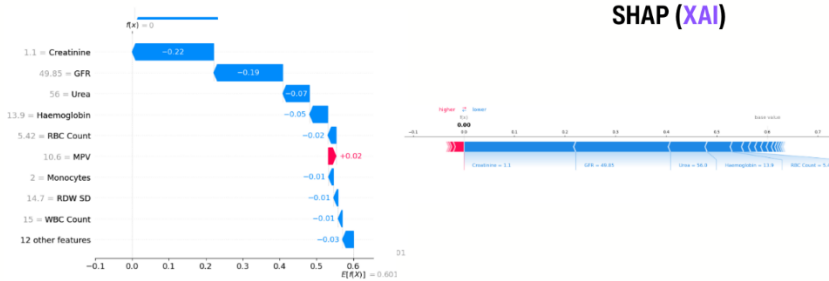


Yes, the patient has CKD.

Reason: The serum creatinine is significantly elevated at 10.9 mg/dL, and the glomerular filtration rate (GFR) is very low at 4.05 mL/min/1.73m². These values strongly indicate a severe decline in kidney function, consistent with Chronic Kidney Disease (CKD). The blood urea is also elevated, further supporting this diagnosis.

Input : {'Sex': 0, 'Age': 65.0, ..., 'Creatinine': 1.1, 'Urea': 56.0, 'GFR': 49.85} (Not CKD Case)

LLM Rationale



No, the patient does not have CKD.

Reason: The serum creatinine is 1.1 mg/dL and the glomerular filtration rate (GFR) is 49.85 mL/min/1.73m². These values are within the normal range, indicating adequate kidney function. While the white blood cell count is slightly elevated, the creatinine and GFR are the most important indicators for CKD.

Discussion :

Conclusion :

This research proposed an end-to-end CKD prediction framework that fine-tunes a medical LLM on serialized clinical tabular data using LoRA/QLoRA via Unsloth, enabling efficient training and strong classification performance across multiple datasets. Beyond prediction, the model generates patient-specific natural language rationales, offering a more accessible form of interpretability compared to technical post-hoc plots. The results support the feasibility of combining prediction and explanation in a single fine-tuned LLM, motivating future work on prospective validation, template robustness, and faithfulness-aware rationalization.

References :

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